Candidate Seat No: __________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Computer Applications (BCA)

Semester-II University Examination May-2018

CA116 – Numerical and Statistical Methods

Date: 07/05/2018, Monday **Time:10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. **Attempt the two sections in separate answer books.**
2. **Figure to the right indicate marks.**
3. **Make suitable assumptions wherever necessary and state them.**

SECTION-I

Q.1 Do as directed : **[11]**

1. What is the probability of getting a 5 from throws of a dice?
   A. 3/6 **B.** 1/6 **C.** 4/6 **D.** 5/6
2. Find out Mode for scores: 1,2,3,2,3,1,2.
3. _____ is used to denote the process of finding the values inside the interval (X_0, Xn).
   A. Interpolation B. Extrapolation C. Iterative D. Polynomial equation.
4. Write down a sample space for rolling a dice and then tossing a coin.
5. Find mean for marks: 10,15,2,3,5.
6. Which operation can be used in Gauss Jordan Method?
   A. Elementary row operations **B.** Multiplication
   C. Addition **D.** Elementary Column operations.
7. Find value for $\mathbf{10C_4}$**.**
8. In Newton-Raphson Method, which formula is used to find out first approximation value?
9. In ______, we must arrange the data before calculating.
   A. Mean **B**. Median **C**. Mode **D**. Range
10. The backward difference interpolation operation is denoted by the symbol_______.
11. Which are correct initial values of $f(x) = x^2-3x-3$ to find the root of nonlinear equation?

Q.2[A] A survey regarding the heights (in cm) of 51 girls of Class X of a school was conducted and the following data was obtained: Find the median height. **[7]**

Height(in cm)	**Number of girls**
Less than 140	4
Less than 145	11
Less than 150	29
Less than 155	40
Less than 160	46
Less than 165	51

[B] Consider families with 4 children. What is the probability of having: **[5]**

1. two boys and two girls
2. at most two boys
3. no boys

OR

Q.2[A] List all types of event in probability and explain any three event with example. **[7]**

[B] **A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarized it in the table given below. Find the mode of the data.** **[5]**

Number of cars	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	7	14	13	12	20	11	15	8

Q.3[A] **Do as directed:** **[6]**

1. The wickets taken by a bowler in 10 cricket matches are as follows:
   2,6,4,5,0,2,1,3,2,3
   Find the **Mode** of the data.
2. Find the **Median** of the data.
   53,62,30,30,34
3. Find the **Mean** of the data.
   17,27,26,2,1,21,13,16,19,5,7

[B] **There are 13 men and 10 women in a running club. A committee of 3 men and 3 women to be selected:** **[6]**

1. How many ways can the 3 men selected?
2. How many ways can the whole committee be selected?
3. A random sample of 6 people is chosen. Find the probability it consist of 3 men and 3 women.

OR

Q.3[A] **For a selected group of people, an insurance company recorded the following data. Find Mean and Mode.** **[6]**

Age	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Number of death	2	12	55	95	71	42	16	7

[B] **Write sample space for following event (with figure).** **[3]**
A coin is tossed twice. If the second throw results in a head ,a dies is thrown, otherwise a coin is tossed.

[C] **The probability that a bomb dropped from a plane will strike the target is 1/5. If six bombs are dropped, find the probability that:** **[3]**

1. Exactly two will strike the target
2. At least two will strike the target

SECTION-II

Q.4[A] **Do as directed:** **[8]**

1. In Bisection method, if the value of f(c) is negative, then root lies between b & c. **[True/False]**
2. Which one is a suitable method to compute f(0.44)? if x=0.30, 0.33, 0.36, 0.39, 0.42, 0.45?
   A. Both C &D
   B. Lagrange Interpolation Formula
   C. Newton's Backward Difference Interpolation Formula
   D. Newton's Forward Difference Interpolation Formula
3. Which interpolation formula method is suitable to find f (3.75) If x=1,3,4,6,9
4. Differentiate REF and RREF.

5. If median=2 and mean=4 then find mode.
6. If P(A)=3/7, find the probability against the event.
7. In which of the following methods, proper choice of initial value is very important?
   A. Newton-Raphson Method **B**. Bisection Method
   C. Iterative Method **D**. Regula Falsi Method
8. The probability of getting a sum 9 from two throws of a dice is 9/12. **[True/False]**

[B] Construct the difference table for $f(x)=(x+2)^2$ for x=1,2,3,4 and **find $\nabla^2 f(3)$.** **[3]**

Q.5[A] Evaluate $f(x) = x^3-3x-5$ using Regula Falsi Method up to four steps. **[7]**

[B] **Evaluate the values of f(2) using Lagrange Interpolation formula for the table of values given below.** **[5]**

x	1.2	2.5	4	5.1	6	6.5
F(x)	6.84	14.25	27	39.21	51	58.25

OR

Q.5[A] Find a root of $x^2 - 3 = 0$ using Bisection Method. **[7]**

[B] Find the positive root of $x^3-2x-5=0$ using Newton-Raphson Method correct up to 3 decimal places with initial value 1. **[5]**

Q.6 **Answer the following.(Any three)** **[12]**

[A] Solve following equations using Gauss Elimination Method.

$$2y+z= -8$$
$$x-2y-3z= 0$$
$$-x+y+2z=3$$

[B] Solve following equations using Gauss Jordan Method.

$$x+3y+z=10$$
$$x-2y-z= -6$$
$$2x+y+2z=10$$

[C] **Compute f (7.5)** by using Newton Backward Interpolation method on following data.

X	5	6	7	8
f(x)	126	217	344	513

[D] **Compute f (0.4)** by using Newton Forward Interpolation method on following data.

X	0.3	0.5	0.6
f(x)	0.61	0.69	0.72

[E] Convert following matrix into RREF.

$$\begin{matrix} 1 & 2 & 0 \\ 3 & 8 & 1 \\ 2 & 4 & 6 \end{matrix}$$